Sparse Cross-Document Contextualization for Persistent Native Memory

An Oracle Headroom Study beyond Packed RAG

Jorge Simão

Draft — September 2026

Abstract

Retrieval-augmented generation (RAG) normally serializes selected documents into one causal token sequence, so later documents are contextualized by earlier documents before the query is decoded. Progressive Retrieval Attention (PRA) instead allows selected resources to remain independently addressable and to persist as native key/value memory. Prior work established that contiguous PRA transport can preserve model behavior exactly, that pre-RoPE keys can be rebound to request-specific packed positions, and that independently stored resources closely reproduce a packed computation when document-to-document attention is disabled. The remaining quality gap is therefore primarily a cross-document-contextualization problem rather than a position-transport problem.

This paper asks whether the useful portion of packed-RAG cross-document interaction is sparse. We instrument the unchanged host-model attention path, extract an oracle graph over physical layer/head/token edges, and replay the same frozen retrieval selection under exact sparse masks. Candidate identities, record order, prompt semantics, model revision, and generation settings remain fixed. Compressed graphs retain only causal cross-record edges; immutable selection receipts and graph digests tie every intervention to its source.

The inception experiment evaluates edge-count budgets from 0 to 100% and cumulative teacher-mass targets from 50 to 99% on MultiHop-RAG. It also freezes disjoint 2,000/150/150 train/validation/test identities for a possible learned selector. On a ten-question mechanism cohort, 0.1% of physical edges reached .1488 F1 versus .1490 for packed RAG and .1431 for independent PRA, but the 0.0059 packed–independent gap is small and the cohort's packed official score was only .600. Larger edge and cumulative-mass budgets were non-monotonic. The prespecified .19/.67 gate therefore remains closed and no selector is trained. A disjoint ten-question validation study compares raw attention with document-pair and layer leave-one-group-out NLL and first-step JS rankings. Its packed–independent direction reverses and every frontier remains non-monotonic. The result shows possible sparse headroom while rejecting both raw attention mass and these coarse causal proxies as sufficient utility signals.

1 Introduction

RAG systems solve a retrieval problem and a context-construction problem. Retrieval selects useful evidence. Context construction then decides how that evidence is presented to the language model. The common implementation packs retrieved text into a causal sequence,

$$D_1\, D_2\, \cdots\, D_m\, Q,$$

and freshly prefills the whole sequence. This creates a dense and order-dependent computation graph: D_2 can attend to D_1, D_3 can attend to both D_1 and D_2, and the final query can attend to all preceding states.

PRA exposes a different design point. Each D_i can remain a persistent addressable record with reusable model-native K/V. The query can later select and materialize those records without rebuilding all source states. The advantage is straightforward when a selection repeats or when records recur across changing queries. The harder question is composition: independent records do not contain the cross-document causal history introduced by ordinary packed RAG.

Paper 3.2 localized this difference. Contiguous selected-context transport was behaviorally exact across multiple model scales and families. Pre-RoPE request-time rebinding reproduced the positional geometry of a packed request, and independently encoded records closely tracked a packed control in which document-to-document attention was disabled. In the powered natural comparison, however, packed RAG remained stronger than independent PRA. A small trained residual recovered part of the gap but did not establish a qualified solution. Fixed sparse masks were non-monotonic, while one very small boundary interaction budget approached packed F1 on some metrics.

These results motivate a narrower question:

> **How much cross-document interaction is actually necessary, and can the necessary interactions be predicted and executed sparsely at request time?**

We therefore separate three quantities:

1. **persistent source state**, stored once as reusable PRA records;
2. **request-specific interaction selection**, which determines which records/tokens/layers need contextualization;
3. **request-specific interaction execution**, which applies a small amount of the frozen base model's own computation.

The intended architecture is

$$\text{persistent PRA K/V} \rightarrow \text{sparse interaction selector}$$
$$\rightarrow \text{small request-local cross-record computation} \rightarrow Q.$$

Contributions. This inception study makes four contributions:

1. A host-preserving attention observer and sparse-mask replay path that leaves base-model parameters and persistent records unchanged.
2. A compressed physical-edge graph and immutable interaction-plan receipt spanning layers, heads, record pairs, and token positions.
3. A prespecified oracle sparsification study over edge-count and cumulative-attention-mass budgets, with task and distribution metrics.
4. A leakage-safe data split and an explicit learned pair-selector design that remains locked unless oracle evidence justifies training.

2 Related Work

RAG and retrieval-conditioned language models place external evidence in the model context or retrieve it during generation [5, 4, 3]. Sparse-attention systems reduce long-sequence cost through fixed, random, global, or learned access patterns [1, 10, 7]. Other work retrieves external states at attention time or restricts K/V traffic [2, 8, 6]. These approaches motivate sparse access, but the present question is different: the retrieved records are already selected and persist as independently encoded native memory. We ask which request-specific cross-record interactions must be restored to recover the behavior of a conventional packed prefix. CacheBlend also studies reuse when separately cached chunks need request-time fusion [9]; our unit of intervention is an auditable physical attention edge under a frozen RAG selection.

3 Relationship to Prior PRA Work

3.1 Paper 1.5: request-time RoPE binding

Paper 1.5 separated coordinate correctness from causal contextualization and showed that, for RoPE models,

$$K_{\text{post}}(p) = R_p K_{\text{pre}}.$$

Pre-RoPE storage therefore allows a persistent key to be rebound to a request-specific effective position without re-encoding the source.

Paper 3.3 assumes that positional contract and does not reopen the positional transport question except as a runtime qualification requirement.

3.2 Paper 3.0: materialization is a computational choice

Paper 3.0 showed that more materialized K/V is not automatically better and that useful detail depends on the query, evidence distribution, and consumer layer. This motivates a query-conditioned interaction policy rather than a fixed global cross-document mask.

3.3 Paper 3.2: causal localization of the composition gap

Paper 3.2 supplies the direct starting point. With a strong frozen selector, packed RAG reached approximately .199 token F1 and .700 official score on the final held-out comparison, while independent PRA reached approximately .135 and .433. A rank-8 request-local residual raised PRA to approximately .154 F1 and .533 official score but left substantial headroom and an uncertain five-seed F1 effect.

More importantly, Paper 3.2 compared

$$A = \text{full causal packed RAG},$$

$$B = \text{packed RAG with document-to-document attention disabled},$$

and

$$C = \text{independent pre-RoPE PRA records rebound to the same positions}.$$

B and C were behaviorally close, while a shape-matched storage/rebinding control was exact across the tested precision modes. The principal semantic difference is therefore cross-document causal interaction.

Paper 3.3 does not attempt to make independent records reproduce every dense packed state. It asks which subset of those interactions is needed for task quality.

4 Problem Formulation

Let the selected evidence consist of m persistent records

$$\mathcal{D} = \{D_1, \ldots, D_m\},$$

with total token count T. Let $\mathcal{E}_{\text{full}}$ denote all document-to-document attention edges allowed by ordinary packed causal RAG.

We seek a request-specific sparse edge set

$$\mathcal{E}_q \subset \mathcal{E}_{\text{full}}$$

such that

$$|\mathcal{E}_q| \ll |\mathcal{E}_{\text{full}}|$$

while preserving task utility:

$$U(\mathcal{D}, Q, \mathcal{E}_q) \approx U(\mathcal{D}, Q, \mathcal{E}_{\text{full}}).$$

The system must additionally preserve most persistent native state:

$$C_{\text{new}}(\mathcal{E}_q) \ll C_{\text{prefill}}(\mathcal{D}).$$

Our primary quality target is to recover at least 80% of the independent-to-packed RAG gap under a request-specific interaction budget of at most 10% of dense cross-document interaction.

5 Oracle Sparse Interaction

5.1 Teacher graph extraction

We instrument packed RAG with frozen retrieval, reranking, selected records, order, token budget, model, and generation settings. For every causal document-to-document attention interaction we record sufficient metadata to construct a sparse teacher graph:

- layer and head;
- source and target record/chunk;
- source and target token positions;
- attention mass;
- distance to document boundaries;
- retrieval rank and score.

Let E be the set of causal token pairs whose source and target belong to different records. The stored score array has shape $[L, H, |E|]$, not $[L, H, T, T]$. Source and target token indices and record identities are stored once, so within-record and causally impossible entries never enter the artifact. For each layer, the observer computes probabilities on a diagnostic side path; the hidden-state update still uses MLX's host scaled-dot-product-attention kernel. A parity receipt checks the observer path against ordinary host execution before the graph is interpreted.

5.2 Oracle budgets

We first retain the highest-attention edges at budgets

$$0,\ 0.01,\ 0.05,\ 0.1,\ 0.5,\ 1,\ 2,\ 5,\ 10,\ 100\%.$$

A complementary cumulative-mass condition retains the smallest edge set covering

$$50,\ 75,\ 90,\ 95,\ 99\%$$

of cross-document attention mass.

For replay, a plan contains a Boolean array of shape $[L, H, |E|]$. At layer ℓ, selected physical edges are added to a packed mask that otherwise blocks cross-record attention. The frozen host model then executes the modified mask. The 100% plan must reproduce the packed teacher exactly; the 0% plan must reproduce the blocked packed control. These endpoint checks distinguish a sparsification result from an instrumentation or transport error.

5.3 Interventional ranking targets

Attention magnitude is observational: a large coefficient shows that an interaction occurred, not that it improved the answer. We therefore add a leave-one-group-out utility for a group g,

$$u_{\mathrm{NLL}}(g) = \mathcal{L}_{\mathrm{answer}}(M_{\setminus g}) - \mathcal{L}_{\mathrm{answer}}(M),$$

where M is the packed teacher and $M_{\setminus g}$ removes only the group's cross-document edges. A positive value means that removing the group raises gold-answer NLL. We also measure first-step Jensen–Shannon divergence, $u_{\mathrm{JS}}(g)$, between $M_{\setminus g}$ and M.

The hierarchy is document pair $\to$ layer $\to$ layer/head $\to$ token edge. At each tested level, groups are ordered by measured utility and physical edges within a group are refined by teacher attention. A second pair-level proxy ranks by

$$\max(0, u_{\mathrm{NLL}}(g))\, a_e,$$

where a_e is the teacher attention of edge e in group g. This is a finite-difference-times-attention diagnostic, not a gradient attribution. These rankings are compared with raw attention at matched physical-edge budgets $0, .01, .05, .1, .5$, and 1%; 100% remains an endpoint parity check.

Every natural question is sampled only after restricting the dataset to a named frozen Paper 3.3 split. Atomic per-question checkpoints bind the dataset, model, reranker, split digest, ranking targets, and budgets, so a resumed run cannot mix configurations. A ten-question validation diagnostic localizes the .1% plans by layer, head, and document pair and selects ranking targets. The powered confirmation then requires at least 100 previously untouched test questions. Learned selection remains locked unless an interventional test ranking meets the absolute quality gate and its F1 frontier is non-decreasing through 1%.

5.4 Oracle gate

The learned-selector program proceeds only if a sparse oracle can approach packed-RAG quality. The initial gate is

$$\mathrm{F1} \geq .19, \qquad \text{official} \geq .67,$$

or statistical indistinguishability from packed RAG while using no more than 5% of dense cross-document interactions.

6 Where Useful Interactions Live

We analyze oracle-selected interactions by:

- transformer layer;
- attention head;
- source/target retrieval rank;
- document pair;
- chunk pair;
- token boundary distance;
- query overlap;
- entity/token overlap;
- gist similarity.

This analysis determines the appropriate routing granularity. If useful interactions are largely pair-level, we avoid token-level prediction. If they concentrate in a few layers or heads, request-local compute can be reduced further.

7 Learned Interaction Selection

This section specifies the next-stage design; it is not evaluated in the present paper. Training is deliberately locked by the failed inception gate.

7.1 Pair selector

The first learned policy predicts interaction importance for selected pair (D_i, D_j):

$$s_{ij} = f_\theta(g_q, g_i, g_j, r_i, r_j, \phi_{ij}),$$

where g_q is a query gist, g_i, g_j are persistent record/chunk gists, r_i, r_j are retrieval/reranker features, and ϕ_{ij} contains pairwise semantic features.

The pair selector is trained against oracle pair importance while controlling a request-specific interaction budget.

7.2 Layer/head selector

If the oracle is concentrated, we extend the policy to

$$s_{ij\ell h}$$

or a factored approximation.

The paper prefers the simplest selector that reaches the quality/economics gate.

8 Sparse Request-Local Contextualization

Synthetic parameter-free K/V append failed in Paper 3.2. The present paper therefore prioritizes on-manifold computation using the frozen base transformer's own projections and attention operations.

Candidate mechanisms are:

Sparse packed-mask composition. Use selected interactions in a request-local composition pass as the first oracle/runtime control.

Boundary interaction. Permit selected real boundary tokens to interact across selected record pairs.

Selected-token interaction. Use a learned token gate within selected record pairs.

Selected-layer propagation. Apply interaction only at selected layers, then propagate corrected request states through the remaining decoder.

Persistent record K/V remains immutable. All corrections are request-local.

9 Training Objective

The base model remains frozen initially.

We optimize

$$\mathcal{L} = \lambda_{\text{task}}\mathcal{L}_{\text{answer}} + \lambda_{\text{seq}}KL(P_{\text{sparse}}\|P_{\text{packed}}) + \lambda_{\text{sel}}\mathcal{L}_{\text{selector}} + \lambda_{\text{cost}}C_{\text{interaction}}.$$

Task quality receives the dominant weight. Global K/V reconstruction is not a primary objective because Paper 3.2 showed that state proximity, answer NLL, and generated-answer quality need not agree.

10 Order Robustness

Packed RAG changes its causal graph when selected documents are permuted. PRA records offer the possibility of more stable set-like evidence composition.

We evaluate canonical, reverse, and random permutations under:

- packed RAG;
- independent PRA;
- oracle sparse composition;
- learned sparse composition.

We report task quality together with mean pairwise JS, unique outputs per question, and F1/NLL variance. Lower order sensitivity is considered a benefit only at matched or improved task quality.

11 Experimental Protocol

11.1 Dataset

MultiHop-RAG is the first mandatory dataset because Paper 3.2 already provides frozen retrieval, canonical records, support labels, and powered natural baselines.

Additional datasets are introduced only after the mechanism passes the primary gate.

11.2 Selection

Use the strongest qualified matched selector from Paper 3.2:

$$\text{BM25} \rightarrow \text{BGE-v2-M3}.$$

Candidate and selection receipts are frozen before composition conditions run.

11.3 Training split

The frozen split uses:

- 2,000 training questions;
- 150 validation questions;
- 150 fixed held-out test questions evaluated with seeds 11, 23, 37, 71, and 101.

The split seed is 3301. All 30 identities declared by the final Paper 3.2 rank-8 residual manifest are excluded from every Paper 3.3 split, rather than only from training. The dataset revision, corpus revision, exact identities, legacy manifest, and split digest are persisted. Oracle smoke questions do not train or select parameters.

11.4 Model ladder

M0 Qwen3-1.7B: full discovery/training matrix.

M1 Qwen3-4B: reduced qualification.

M2 Qwen3-8B: reduced qualification.

M3 Llama-3.1-8B: cross-family qualification.

12 Baselines

Mandatory baselines:

1. packed RAG;
2. independent pre-RoPE PRA;
3. Paper 3.2 rank-8 residual;
4. fixed sparse boundary mask;
5. oracle sparse interaction;
6. learned sparse selector/composer.

13 Metrics

Task quality. Official score, token F1, exact match, gold-answer NLL, and answer log-probability.

Distributional fidelity. First-step JS/KL, logit differences, and generated-output agreement.

Sparsity. Selected document pairs, layers, heads, tokens, and total cross-document interaction edges.

Systems. Persistent K/V reuse, newly computed states, request-local FLOPs, composition latency, TTFT, total latency, and memory.

Robustness. Five-seed intervals, order sensitivity, model-scale transfer, family transfer, and precision sensitivity.

14 Inception Results

14.1 Scope and endpoint validity

We ran the complete oracle grid on ten deterministically selected MultiHop-RAG questions with Qwen3-1.7B-4bit on an Apple M4 Pro with 48 GiB unified memory. BM25 produced 20 candidates, BGE-v2-M3 reranked them, and a 512-token budget retained three or four records. The model, reranker, corpus, and dataset revisions are pinned in the manifest. This is a one-seed mechanism cohort, not a population estimate. A post-freeze provenance check places all ten identities in the Paper 3.3 training partition, with no overlap with the Paper 3.2 final evaluation identities. The validation diagnostic and powered test confirmation therefore remain disjoint from this inception result.

Observer instrumentation matched ordinary host execution on all 10 questions. The 100% physical-edge plan also matched the instrumented packed teacher on all 10, with zero K/V delta and zero first-step JS. The 0% plan matched the packed no-cross-document mask. The average teacher graph contained 38.04 million physical cross-document edges over 28 layers and 16 heads. These checks close the implementation gate: differences inside the sweep are caused by the edge mask, not by an alternate attention kernel.

14.2 Quality–sparsity frontier

At 0.1% physical edges, the oracle recovers 97.3% of this ten-question cohort's small 0.0059 F1 packed–independent gap: F1 is .1488 versus .1490 for packed RAG and .1431 for independent PRA. Official score remained .600 in all three conditions, so no official-score gap existed to recover. The 0.1% plan did not reproduce the same strings as packed RAG: output agreement was 0/10 and first-step JS remained .110. The result is therefore task-level headroom, not behavioral equivalence, and the relative percentage should not be read as a large absolute quality gain.

The frontier is strongly non-monotonic. Increasing the top-edge budget from 0.1% to 0.5% reduced F1 from .1488 to .0968 even though retained teacher mass rose from 36.2% to 80.3%. Likewise, the smallest set carrying 50% of teacher mass used only .149% of physical edges but did not outperform independent PRA. Raw attention mass is thus an inadequate causal-utility target by itself.

Table 1: Paper 3.3 oracle smoke. “Edges” is the fraction of physical layer/head/token-pair cross-document edges. Attention mass is measured under the packed teacher.

Condition	F1	Official	Gold NLL	Edges	Mass
Packed host / oracle 100%	.1490	.600	4.780	100%	100%
Independent PRA	.1431	.600	2.550	0%	0%
Oracle top 0.01%	.1449	.600	2.579	.01%	4.0%
Oracle top 0.05%	.1416	.600	2.571	.05%	19.2%
Oracle top 0.1%	**.1488**	**.600**	2.610	.1%	36.2%
Oracle top 0.5%	.0968	.400	4.794	.5%	80.3%
Oracle top 5%	.0990	.400	4.851	5%	91.9%
Oracle top 10%	.1212	.500	4.852	10%	95.1%
Mass oracle 50%	.1403	.500	2.753	.149%	50%
Mass oracle 75%	.1145	.500	3.757	.324%	75%
Mass oracle 90%	.0768	.400	4.758	3.466%	90%
Mass oracle 99%	.1308	.500	4.855	30.902%	99%

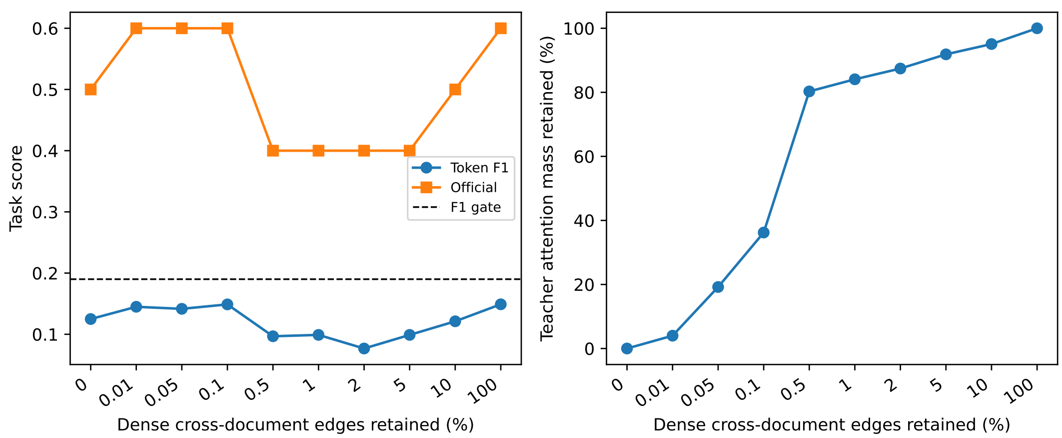

Figure 1: Task quality and retained teacher mass across physical top-edge budgets. The curve is a mechanism smoke over ten questions; connecting lines do not imply monotonicity or statistical precision.

14.3 Interaction localization

Cross-document attention mass was distributed across the network: layers 0–8 contained 27.4%, layers 9–18 contained 31.7%, and layers 19–27 contained 40.8%. No single layer exceeded 4.85% on average, and the largest layer/head cell averaged only .357%. The graph is therefore late-leaning but not localized to one layer or head. Pair mass was more concentrated: the first record supplied 49.1% of mass into the second and 38.8% into the third. These values partly reflect causal geometry and token counts; they are routing features to test, not evidence that early-ranked records are causally sufficient.

14.4 Interventional validation

We next ran the six ranking targets on ten frozen validation questions, using the same model, selector, 512-token budget, and physical-edge replay. The BGE reranker ran on MPS; this execution

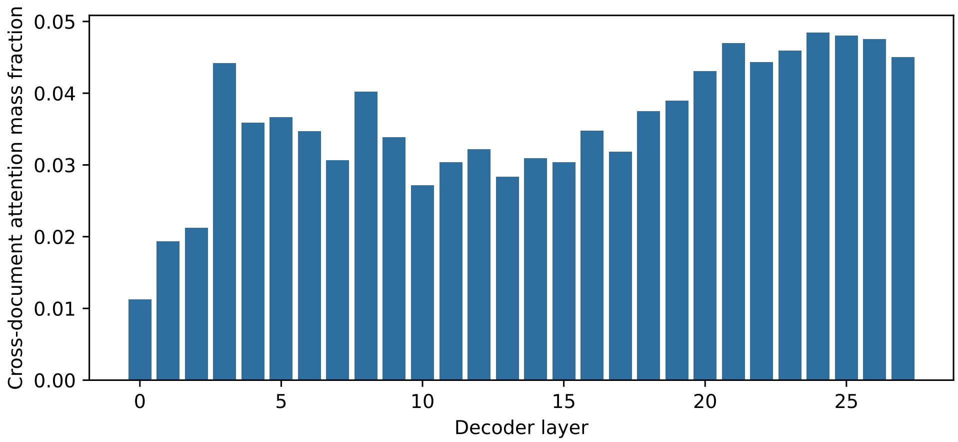

Figure 2: Mean share of packed-teacher cross-document attention mass by decoder layer. Later layers carry more aggregate mass, but no narrow layer band dominates.

device is part of the run configuration and receipt. All ten ordinary/instrumented host comparisons and all ten 100% replay checks passed exactly.

Table 2: Frozen validation target selection. Each oracle row reports its highest-F1 budget at or below 1%; it is not a test-set estimate.

Condition	Edges	F1	Official	Gold NLL
Packed teacher	100%	.1094	.600	5.911
Independent PRA	0%	.1673	.500	3.211
Raw attention	.1%	.1706	.500	3.697
Pair NLL	1%	.1659	.600	3.929
Pair JS	.5%	**.1727**	.600	3.666
Pair NLL $\times$ attention	.1%	.1379	.400	3.861
Layer NLL	.01%	.1540	.400	3.204
Layer JS	.5%	.1525	.500	3.739

This cohort reverses the inception endpoint: independent PRA exceeds packed F1 by .0579, although packed retains .100 more official score. It therefore does not define a packed advantage that sparse edges can meaningfully recover. The apparent improvements are also uncertain. Relative to the packed teacher, pair-JS at .5% changes F1 by +.0633 (paired bootstrap 95% interval $[-.0067, .1618]$) with zero mean official-score change; raw attention at .1% changes F1 by +.0613 ($[-.0275, .1604]$) while changing official score by $-.100$. Both intervals include zero. Gold NLL ranks conditions differently from generated-answer metrics, so lower teacher-forced loss is not treated as proof of better task behavior.

No ranking has a non-decreasing F1 frontier through 1%. The result therefore strengthens, rather than resolves, the inception warning: raw magnitude is not a causal utility signal, and coarse leave-one-group-out NLL or first-step JS is not sufficient by itself. The powered test carries raw attention, pair NLL, pair JS, and pair-NLL-times-attention forward. Layer-only policies are dropped after validation because they do not improve the quality frontier.

The .1% plans also reveal different geometries. Raw attention allocates 32.4% of selected edges to layers 23–25 and 27, while pair NLL concentrates more in early layers and assigns 70.8% to record-index pairs $(1, 2)$ and $(0, 1)$. Pair JS assigns 79.6% to pair $(0, 1)$ but remains distributed

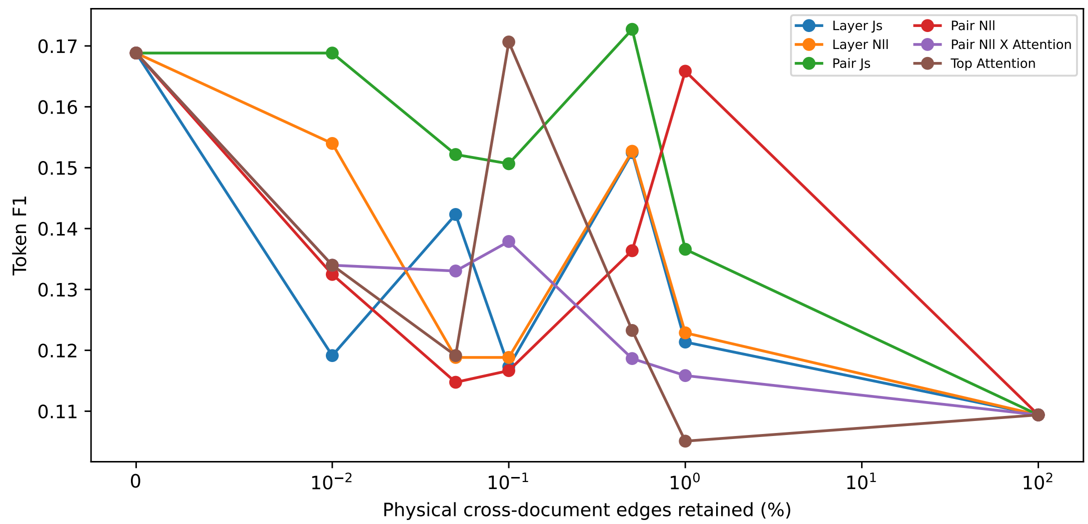

Figure 3: Validation F1 frontiers for observational and interventional rankings. All curves are non-monotonic; lines aid reading and do not imply interpolation.

across layers; the combined proxy again leans late. These are policy-induced concentrations, not discovered universal heads. They nevertheless show that the ranking target changes the proposed sparse computation substantially.

14.5 Go/no-go decision

The inception gate is **not passed**. The 0.1% oracle is close to this cohort's packed F1, but neither condition reaches the prespecified absolute thresholds of .19 F1 and .67 official score. With one ten-question cohort we also cannot establish statistical equivalence. Learned-selector training therefore remains locked. The validation experiment also finds no monotonic attention-, pair-, or layer-ranked frontier. A 150-question frozen-test oracle is consequently the next decision point; selector learning remains disabled while that preregistered confirmation runs.

The reported encoding times, roughly 0.5–0.9 seconds per condition, execute a dense host attention kernel under a sparse mask. Edge fraction is therefore a logical interaction budget, not a measured wall-time or FLOP reduction. A production sparse kernel is downstream work and should be evaluated only after the quality gate passes.

15 Success Criteria

The predeclared practical gate is

$$\text{F1} \geq .19, \qquad \text{official} \geq .67,$$

or statistical indistinguishability from packed RAG, while using no more than 5% of dense request-specific cross-document interaction. Approximate 80% gap recovery remains a useful descriptive quantity, but does not replace these absolute thresholds.

The ten-question smoke does not meet these absolute criteria. Its 0.1% point meets the relative F1-recovery target only because the cohort's packed and independent endpoints are close. We do not substitute that post-hoc relative observation for the prespecified gate.

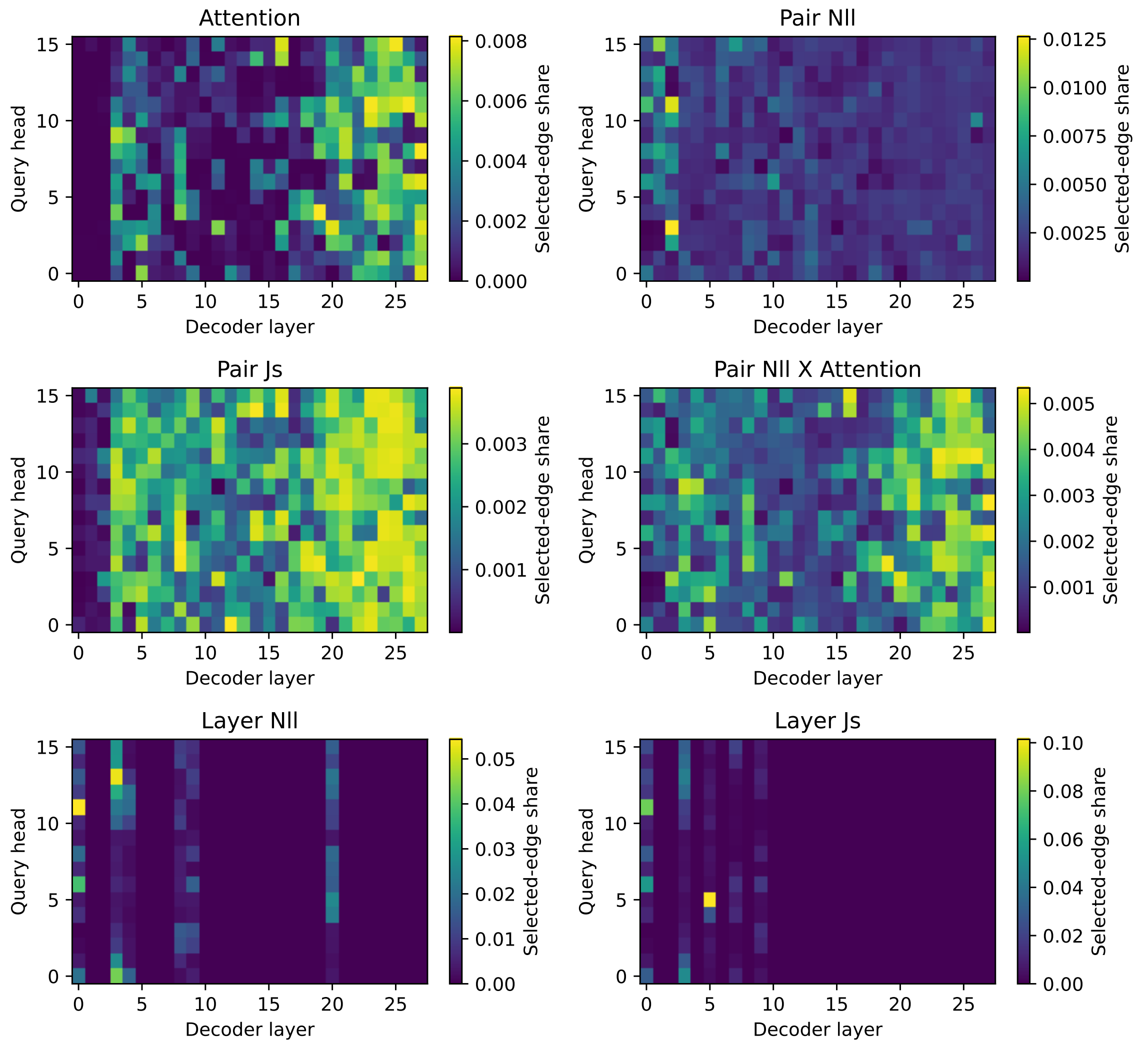

Figure 4: Layer–head allocation of each validation .1% plan, aggregated over ten frozen validation questions. Color is selected-edge share within a policy.

A stronger result matches packed RAG within uncertainty while preserving:

- persistent native K/V reuse;
- lower order sensitivity;
- materially lower request-specific compute.

16 Falsification

Scientifically useful negative outcomes include:

- an oracle requires dense interaction;
- learned selection fails to approach the oracle;

- sparse state propagation destroys quality;
- selector overhead erases saved prefill;
- order robustness disappears at matched quality;
- the mechanism does not transfer across models/families.

17 Systems Integration

Paper 3.3 reuses the canonical Paper 4.5/Paper 3.2 `ContextRecord` and storage metadata. The learned interaction policy is request-local and must not mutate persistent source records.

The target runtime abstraction is

$$\texttt{ContextRecord} \to \texttt{SelectionReceipt} \to \texttt{InteractionPlan} \to \texttt{RequestLocalComposition}.$$

An `InteractionPlan` should record selected record pairs, token spans, layers/heads, budget, policy revision, and model/runtime compatibility.

18 Discussion

The 0.1% result suggests that conventional RAG may perform substantially redundant document-to-document computation. It does not yet establish that claim: the sample is small, the absolute gate failed, and the present sparse mask still executes a dense kernel. PRA could eventually retain persistent source state and pay only for a small request-specific neural "join" among selected resources, but that economic claim requires both powered quality and a real sparse execution path.

This would change the role of PRA from a cache/retrieval mechanism into a composable native-memory substrate:

$$\text{persistent records} + \text{sparse request-local contextualization}.$$

If it fails, the result would instead bound how separable transformer source context really is: packed causal contextualization may be too distributed to recover cheaply after independent encoding.

19 Conclusion

Paper 3.2 established that position transport and persistent native memory can be separated from the cross-document causal computation induced by packed RAG. Paper 3.3 targets that remaining computation directly. Exact endpoint checks show that the new observer and physical-edge masks isolate this variable. The ten-question smoke finds a promising 0.1% point but also a sharply non-monotonic frontier, so attention mass alone is not a reliable selector target. A disjoint validation cohort reverses the packed–independent endpoint and finds that pair-NLL, pair-JS, and layer ablations also remain non-monotonic. The prespecified oracle gate remains closed. A powered frozen-test oracle and a more faithful task-aware importance signal are required before learned selection or sparse runtime economics can be claimed.

References

[1] Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

[2] Amanda Bertsch, Uri Alon, Graham Neubig, and Matthew R. Gormley. Unlimiformer: Long-range transformers with unlimited length input. In *Advances in Neural Information Processing Systems*, 2023.

[3] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George B. M. van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning*, 2022.

[4] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. Retrieval augmented language model pre-training. In *International Conference on Machine Learning*, 2020.

[5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, 2020.

[6] Di Liu, Meng Chen, Baotong Lu, Huiqiang Jiang, Zhenhua Han, Qianxi Zhang, Qi Chen, Chengrui Zhang, Bailu Ding, Kai Zhang, Chen Chen, Fan Yang, Yuqing Yang, and Lili Qiu. Retrievalattention: Accelerating long-context LLM inference via vector retrieval. In *Advances in Neural Information Processing Systems*, 2025.

[7] Aurko Roy, Mohammad Saffar, Ashish Vaswani, and David Grangier. Efficient content-based sparse attention with routing transformers. *Transactions of the Association for Computational Linguistics*, 9:53–68, 2021. Preprint: `https://arxiv.org/abs/2003.05997`.

[8] Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. QUEST: Query-aware sparsity for efficient long-context LLM inference. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[9] Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang. CacheBlend: Fast large language model serving for RAG with cached knowledge fusion. *arXiv preprint arXiv:2405.16444*, 2024.

[10] Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. Big bird: Transformers for longer sequences. In *Advances in Neural Information Processing Systems*, 2020.

A Interventional Replay Procedure

For each frozen selection, the implementation performs the following steps:

1. Encode the concatenated records with ordinary full-causal attention, observing only causal cross-record probabilities into a compressed $[L, H, |E|]$ graph.

2. Encode the same tokens with every cross-record edge blocked. This is the base mask to which sparse plans add visibility.
3. For each document-pair or layer group g, replay the full graph except g, score the gold answer without decoding, and record $u_{\mathrm{NLL}}(g)$ and $u_{\mathrm{JS}}(g)$.
4. Order groups by one measured utility. Within a group, order physical edges by teacher attention, or multiply non-negative group NLL utility by edge attention for the combined proxy.
5. Replay nested prefixes at each physical-edge budget. Generate and teacher-force the answer from the same immutable selection and query.
6. Verify that the empty plan matches the blocked control and that the full plan matches the instrumented packed teacher exactly.

The implementation obtains only the prefix needed by the largest sparse budget through deterministic partition selection. It does not sort all tens of millions of edges for each target. This optimization changes oracle-analysis latency, not the selected set or the dense masked host execution used to test quality.

B Artifact Schema

Each run should persist:

- dataset/query IDs and seed;
- candidate and selection receipts;
- persistent record URIs/versions;
- document order;
- teacher interaction graph digest;
- oracle/learned interaction-plan digest;
- leave-one-group-out NLL and first-step JS utility;
- selected pairs/layers/heads/tokens;
- interaction budget;
- training checkpoint/policy revision;
- precision/runtime metadata;
- task/distribution/system metrics;
- paired bootstrap effects against the packed teacher;
- reranker execution device and frozen-split digest;
- git commit and hardware.

C Reproduction Map

The physical graph, plan, and localization implementation is in `src/pra_hf/sparse_crossdoc.py`. The host-preserving observer and sparse mask seam is in `src/pra_hf/rag_mlx_native.py`. The complete frozen retrieval and oracle sweep is in `experiments/paper3_3_sparse_crossdoc/run_oracle_sparsity.py`; split generation is in the adjacent `freeze_splits.py`. Exact commands and evidence labels appear in the experiment and paper README files.

The publication manifest named in the README contains row metrics, receipts, group interventions, selected-edge localization, monotonicity diagnostics, paired intervals, runtime metadata, and plot inputs. Regenerable teacher graphs and per-question recovery checkpoints are excluded from Git; graph content digests and checkpoint-bound configuration digests remain in the final manifest.